

Study on Flame Characteristics and Pollutant Distribution in Methane-Hydrogen Blended Combustion

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Abstract: The environmental pollution resulting from conventional fossil fuel combustion necessitates the exploration of cleaner and lower-carbon energy sources. This research conducts numerical simulations on methane-hydrogen combustion technology in atmospheric burners, utilizing the eddy dissipation model to simulate methane-hydrogen combustion and examining the effects of different hydrogen blending ratios and equivalence ratios on combustion temperature, flame shape, and primary pollutant emission concentrations. Computational results indicate that increases in the hydrogen blending ratio and equivalence ratio both lead to a rise in combustion temperature, subsequently elevating NO_x concentration levels. With the augmentation of the hydrogen blending ratio and equivalence ratio, opposing impacts on flame position and shape are observed; the former predominantly causes flame contraction, while the latter induces flame expansion.

Keywords: Atmospheric Burner, Hydrogen-Blended Combustion, Equivalence Ratio, Numerical Simulation